Notations: $\dot{x} = \frac{dx}{dt}$

$y' = \frac{dy}{dx}$ (usually)

but also $\frac{dy}{dt}$ (context-based)

$\dot{x} = f(x)$ $\qquad$ $\dot{y} = g(t, y)$

find func: $t \mapsto x(t)$ $\qquad$ $t \mapsto y(t)$

dependent variable

$\Rightarrow \frac{dx}{dt} = f(x(t))$ $\qquad$ $\frac{dy}{dt} = g(t, y(t))$

independent variable

We assume that $t \mapsto x(t)$ is differentiable.

In ODE's, the independent variable is one dimensional (usually denoted by t)

Our aim: find whether & when a such eqn has a soln & determine its properties.

Examples: Newton's law:

$$F = ma \quad \text{or} \quad F(x(t)) = m\ddot{x}(t)$$

$\frac{d^2x}{dt}$

gravitational force.

In 3D: $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$

$$m\ddot{x}_1 = 0 \quad , \quad m\ddot{x}_2 = 0 \quad , \quad m\ddot{x}_3 = -g$$

This has a solution

$$x(t) = x(0) + v(0)t - \frac{g}{m}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} t^2$$

Practice! how did you arrive to this?

So, according to Newton's law, gravitational pull btw 2 particles of mass m_1 & m_2 is

$$F(x) = \gamma \frac{m_1 m_2 x}{|x|^3}$$

$$\Rightarrow \quad m\ddot{x}_i = -\frac{\gamma m_1 m_2 x_i}{|x_1^2 + x_2^2 + x_3^2|^{3/2}} \qquad \text{for } i = 1,2,3$$

What about 3 bodies / particles?
we have something similar.

$\Rightarrow$ Now, it becomes difficult to solve this equation.

Chaotic? sol$^{\underline{n}}$ exists? one? many? none?

So how can we tell?

Consider an IVP

$$\frac{d\vec{x}}{dt} = f(t, \vec{x}) \quad \& \quad x(0) = x_0$$

where $f : \mathbb{R} \times \mathbb{R}^n \longrightarrow \mathbb{R}^n$

Does this IVP have sol$^{\underline{n}}$s?
Are the sol$^{\underline{n}}$s unique?

There might NOT exist any sol$^{\underline{n}}$.

Example:

Consider $x' = f(x)$ where

$$f(x) = \begin{cases} 1 & x < 0 \\ -1 & x \geq 0 \end{cases}$$

$\Rightarrow \nexists$ <u>no</u> sol$^{\underline{n}}$ to the initial value

$$\dot{x} = f(x) \qquad x(0) = 0.$$

Question: Is

$$x(t) = \begin{cases} t & \text{for } t \leq 0 \\ -t & \text{for } t > 0 \end{cases}$$

a solution?

Sol$^{\underline{n}}$s may not be unique

Example:

Let $\dot{x} = |x|^{\frac{1}{2}} \qquad x(0) = 0$

$$\Rightarrow \quad x(t) = \begin{cases} 0 & \text{for } t < 1 \\ (t-1)^2/4 & \text{for } t \geq 1 \end{cases}$$

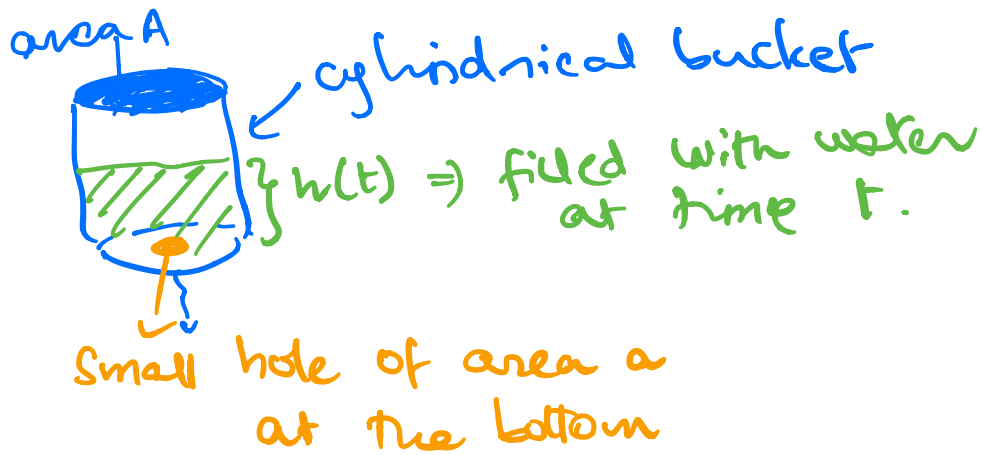
is a sol$^{\underline{n}}$.

Question: What other sol$^{\underline{n}}$s can you have?

How many sol$^{\underline{n}}$s does this IVP have?

Physical meaning of non-uniqueness

Example: (Bucket problem)



Water leaks out of the bucket under the effect of gravity

The eq$^{\underline{n}}$ becomes

$$\dot{h} = \frac{-a}{A}\sqrt{2gh}$$

Sol$^{\underline{n}}$ of this eq$^{\underline{n}}$ is not unique.

Question: Why is non-uniqueness not surprising? (Hint: consider the problem

backward in time)

Having a unique solⁿ of the IVP

$$\frac{dx}{dt} = f(t, x) \quad \& \quad x(0) = x_0$$

means the problem is deterministic

Background on modelling (not included).

Water flowing into the system $Q_{in} = 0$

$$Q_{stored} = -Q_{out}$$

Volumetric flow rate

$$A \frac{dh}{dt} = -a\, v(t)$$

velocity $v(t)$ of water coming out is $\sqrt{2gh}$

$$\Rightarrow \quad A \frac{dh}{dt} = -a\sqrt{2gh}$$

$$\Rightarrow \quad \frac{dh}{dt} = -\frac{a}{A}\sqrt{2gh}$$

Can you think of a method to solve this?

$$\frac{dh}{dt} + \frac{a}{A}\sqrt{2g}\sqrt{h(t)} = 0$$

$$\Rightarrow \frac{dh}{dt} + K\sqrt{h(t)} = 0 \ ?$$

Separation of variables:

$$\int_{h_0}^{h} \frac{dh}{\sqrt{h(t)}} = -\int_0^{t_1} \frac{a\sqrt{2g}}{A}\, dt.$$

$$\Rightarrow 2h^{\frac{1}{2}}\Big|_{h_0}^{h} = -\frac{a\sqrt{2g}}{A}(t_1 - 0)$$

$$\Rightarrow \sqrt{h} - \sqrt{h_0} = -\frac{a\sqrt{2g}}{2A}$$

$$\Rightarrow \sqrt{h} = \sqrt{h_0} - \frac{a\sqrt{2g}}{2A}t_1$$

$$\Rightarrow h = \left(\sqrt{h_0} - \frac{a\sqrt{2g}}{2A}t\right)^2$$

When will the bucket be empty? ie. $h=0$?

$$\sqrt{h_0} - \frac{a\sqrt{2g}}{2A} t = 0$$

$$\Rightarrow \quad t_{empty} = \frac{2A\sqrt{h_0}}{a\sqrt{2g}} = \frac{A\sqrt{2h_0}}{a\sqrt{g}}$$

Is the sol^{n} unique?